

PHILIP SOLIMINE

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EXPERIENCE

Microsoft

Senior Data & Applied Scientist

2026 - Present

Chewy

Data Scientist II

2025-2026

Vancouver School of Economics, University of British Columbia

Postdoctoral Research & Teaching Fellow - Centre for Innovative Data in Economics Research

2022-2025

Departments of Economics and Scientific Computing, Florida State University

Charles & Persis Rockwood Fellow, L. Charles Hilton Fellow

2017-2022

EDUCATION

Florida State University

PhD Economics

July 2022

Dissertation: Economic behavior in dynamic networks

MS Computational Science

July 2022

Thesis: Optimal control for networked moments: Theory and applications to network-based targeting

MS Economics

Dec 2018

BA Mathematics (minor in Physics)

Dec 2016

BS Economics (minor in Computer Science)

Dec 2016

PUBLICATIONS

1. Li, W. and **Solimine, P.** (*forthcoming*). Targeting persistence: Coarse interventions in social networks. *Proceedings of the 27th ACM Conference on Economics and Computation (EC)*. p. 85
2. **Solimine, P.** and Schrimpf, P. (2025). Investment and misallocation in infrastructure networks: The case of U.S. natural gas pipelines. *Proceedings of the 26th ACM Conference on Economics and Computation (EC)*. p. 8.
3. **Solimine, P.** and Isaac, RM. (2023). Reputation and market structure in experimental platforms. *Journal of Economic Behavior & Organization*, 205, 528-559.
4. Dunkle, B., Isaac, RM., and **Solimine, P.** (2022). The robustness of lemons in experimental markets. *Experimental Law and Economics*. Research in Experimental Economics, Vol. 21.
5. **Solimine, P.** and Meyer-Baese, A. (2022). Input design for the optimal control of networked moments. *Proceedings of the 61st IEEE Conference on Decision and Control (CDC)*. 5894-5901.
6. **Solimine, PC.** (2021). Network controllability metrics for corruption research. *Corruption Networks*. Understanding Complex Systems. Springer.
7. **Solimine, PC.** (2020). Political corruption and the congestion of controllability in social networks. *Applied Network Science* (Vol. 5, p. 23).
8. Tahmassebi, A., Mohebbi, B., **Solimine, P.**, Meyer-Baese, U., Pinker, K., & Meyer-Baese, A. (2019, May). Model reduction of structural biological networks by cycle removal. *Proceedings of the SPIE: Smart Biomedical and Physiological Sensor Technology XVI* (Vol. 11020, pp. 105-112).
9. Tahmassebi, A., Mohebbi, B., Meyer-Baese, L., **Solimine, P.**, Pinker, K., & Meyer-Baese, A. (2019, May). Determining driver nodes in dynamic signed biological networks. *Proceedings of the SPIE: Smart Biomedical and Physiological Sensor Technology XVI* (Vol. 11020, pp. 53-60).

WORKING PAPERS

- Investment and misallocation in infrastructure networks (with Paul Schrimpf)
- Coarse targeting in social networks (with Wei Li)
- Strategic formation of collaborative networks (with Luke Boosey) (*submitted*)
- Model recovery and competition in prediction markets

SKILLS AND TECHNICAL EXPERTISE

Programming Languages	Python, C++, Julia, R, SQL, Matlab
Machine Learning	JAX, PyTorch, TensorFlow, PyMC
Big Data Tools	Spark, Hadoop, MPI, CUDA, OpenMP
Specialties	Networks, Machine Learning, AI, Causal Inference, Structural Estimation, Auctions, Experimentation, Industrial Organization, Dynamic Games
Software Tools	Git, Docker, Poetry, VS Code, JIRA

PROJECTS

PyRiesz: An open source PyTorch implementation of the multitasking deep RieszNet architecture for Auto-DML causal effects estimation.

Created and maintained an open-source PyTorch implementation of the multitasking deep RieszNet architecture, and high-dimensional LASSO regression methods to automatically debias machine learning models by learning Riesz representers.

Robustness and regulation in the face of adversarial discord (with Wei Li and Jesse Perla)

Applied control theory to study network robustness and manipulation. Developed tools to counter manipulation through information design and design-based passive moderation in networked platforms.

Pricing network effects with endogenous community structure (with Angelo Mele)

Examined pricing and user engagement dynamics on a digital platform using a large dataset. Estimated demand with dynamic network effects through a network formation game model using variational approximations and mean-field game theory.

Viral dynamics and coordinated promotion in digital platforms (with Matthew Gentry)

Studied the impact of pricing strategies on consumer dynamics and demand in the video game industry. Applied computer vision and data mining to decompose game features. Developed a structural econometric model for demand dynamics in competitive video game markets.

TEACHING

University of British Columbia

- **ECON 323 Quantitative Economic Modeling and Data Science** 2022-2025 ($\times 10$ sections)
Topics covered: Programming fundamentals in Python, Data engineering with Pandas, Data science tools, Applied linear algebra, Numerical methods, Visualization, Machine learning, Network economics
- **ECON 622 Computational Economics (PhD)** 2023
Topics covered: Graphical models, Modern MCMC methods, Gibbs sampling, Probabilistic programming, Frequentist and Bayesian inference, Dynamic discrete choice, Machine learning theory, NLP
- **ECON 526 Quantitative Economics (MA)** 2023
Topics covered: Causal inference, Research design, Uncertainty quantification, Directed graphical models, Experimentation

Florida State University

- **ECO 4400 Games and Decisions** 2020 (online), 2021
Topics covered: Decision theory, Optimization, Decision under risk, Nash equilibrium, Strategy, Industrial organization, Cournot competition, Bertrand competition, Dynamic games, Auctions
- **ECO 2023 Principles of Microeconomics** 2019
Topics covered: Opportunity cost, Marginal cost and marginal benefit, Supply and demand, Revenue and cost curves, Profits and utility, Equilibrium, Introduction to game theory
- **ECO 5434 Analysis of Economic Data (MS) (guest lecturer)** 2022
Topics covered: Social and economic networks

CONFERENCE TALKS & PRESENTATIONS

- **2025:** University of Washington (invited), Bank of Canada (invited), Access to Cash and Financial Services Workshop (invited), ACM EC Conference on Economics and Computing
- **2024:** UBC Sauder Industrial Organization Workshop, Conference of Network Science in Economics, INFORMS
- **2023:** International Industrial Organization Conference; UBC Econometrics Seminar
- **2022:** IEEE Conference on Decision and Control, UBC Econometrics Group (invited); Conference of Network Science in Economics ($\times 2$); FSU Computational Xposition; FSU Quantitative Methods Group; FSU Microeconomic Theory Seminar
- **2021:** Conference of Network Science in Economics; Economic Science Association Job-Market Candidates Seminar; North American Meeting of the Economic Science Association; Networks 2021 (NetSci and Sunbelt); Conference of the Southern Economic Association; FSU Experimental Seminar
- **2020:** NetSci 2020 (invited); Network Science in Economics; Global Meeting of the Economic Science Association; FSU Computational Xposition; FSU Experimental Seminar
- **2019:** Caltech Symposium in Honor of Charles R. Plott (invited); Conference of the Southern Economic Association; NetSci 2019; FSU Experimental Seminar

PROFESSIONAL SERVICE

Referee Reports

- Review of Economics & Statistics
- Journal of Economic Behavior & Organization ($\times 2$)
- International Journal of Industrial Organization ($\times 3$)
- Economics, Applied Network Science, Government & Opposition, Economics Letters